The effect of high-protein diets on coronary blood flow.

Recent research has demonstrated that successful simultaneous treatment of multiple risk factors including cholesterol, triglycerides, homocysteine, lipoprotein (a) [Lp(a)], fibrinogen, antioxidants, endothelial dysfunction, inflammation, infection, and dietary factors can lead to the regression of coronary artery disease and the recovery of viable myocardium. However, preliminary work revealed that a number of individuals enrolled in the original study went on popular high-protein diets in an effort to lose weight. Despite increasing numbers of individuals following high-protein diets, little or no information is currently available regarding the effect of these diets on coronary artery disease and coronary blood flow. Twenty-six people were studied for 1 year by using myocardial perfusion imaging (MPI), echocardiography (ECHO), and serial blood work to evaluate the extent of changes in regional coronary blood flow, regional wall motion abnormalities, and several independent variables known to be important in the development and progression of coronary artery disease. Treatment was based on homocysteine, Lp (a), C-reactive protein (C-RP), triglycerides, total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, very low-density lipoprotein cholesterol, and fibrinogen levels. Each variable was independently treated as previously reported. MPI and ECHO were performed at the beginning and end of the study for each individual. The 16 people (treatment group/TG) studied modified their dietary intake as instructed. Ten additional individuals elected a different dietary regimen consisting of a "high-protein" (high protein group/HPG) diet, which they believed would "improve" their overall health. Patients in the TG demonstrated a reduction in each of the independent variables studied with regression in both the extent and severity of coronary artery disease (CAD) as quantitatively measured by MPI. Recovery of viable myocardium was seen in 43.75% of myocardial segments in these patients, documented with both MPI and ECHO evaluations. Individuals in the HPG showed worsening of their independent variables. Most notably, fibrinogen, Lp (a), and C-RP increased by an average of 14%, 106%, and 61% respectively. Progression of the extent and severity of CAD was documented in each of the vascular territories with an overall cumulative progression of 39.7%. The differences between progression and extension of disease in the HPG and the regression of disease in the TG were statistically ($p<0.001$) significant. Patients following recommended treatment for each of the independent variables were able to regress both the extent and severity of their coronary artery disease (CAD), as well as improve their myocardial wall motion (function) while following the prescribed medical and dietary guidelines. However, individuals receiving the same medical treatment but following a high-protein diet showed a worsening of independent risk factors, in addition to progression of CAD. These results would suggest that high-protein diets may precipitate progression of CAI) through increases in lipid deposition and inflammatory and coagulation pathways.